SENSE MAKING FROM SENSORY DATA

MULTIMODAL MACHINE LEARNING

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SENSE MAKING FROM SINGLE MODALITY / MULTIMODAL SENSORY DATA

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What is modality

Modality refers to the way in which something expressed or perceived.

Raw modality
(closest from sensor)

Abstract modality
(farthest from sensor)



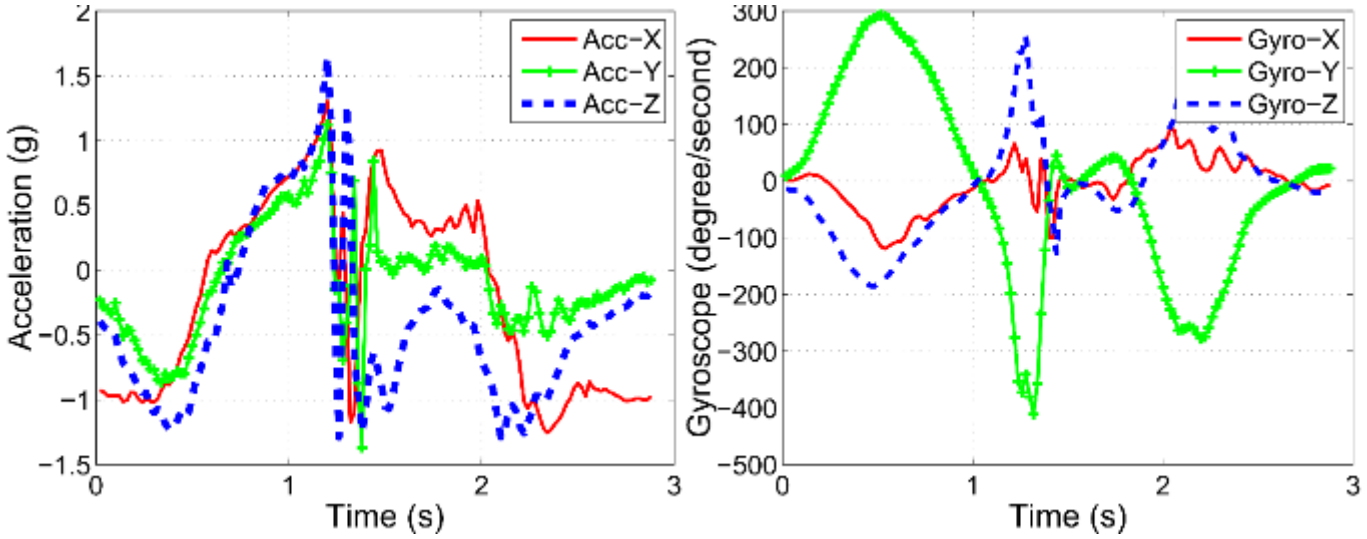
Speech	Language	Sentiment
Image	Detected objects	Object attributes
IoT water sensor reading	Water leakage	Event statistics

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Single modality: Inertial

- An accelerometer essentially measures the force (proper acceleration) along the x, y, and z axes.
- Understanding of position/orientation helps mobile phones reorient the screen in portrait or landscape mode and reverse direction as required.
- Gesture recognition is based on buffering continuous data and looking at the change in force and orientation.

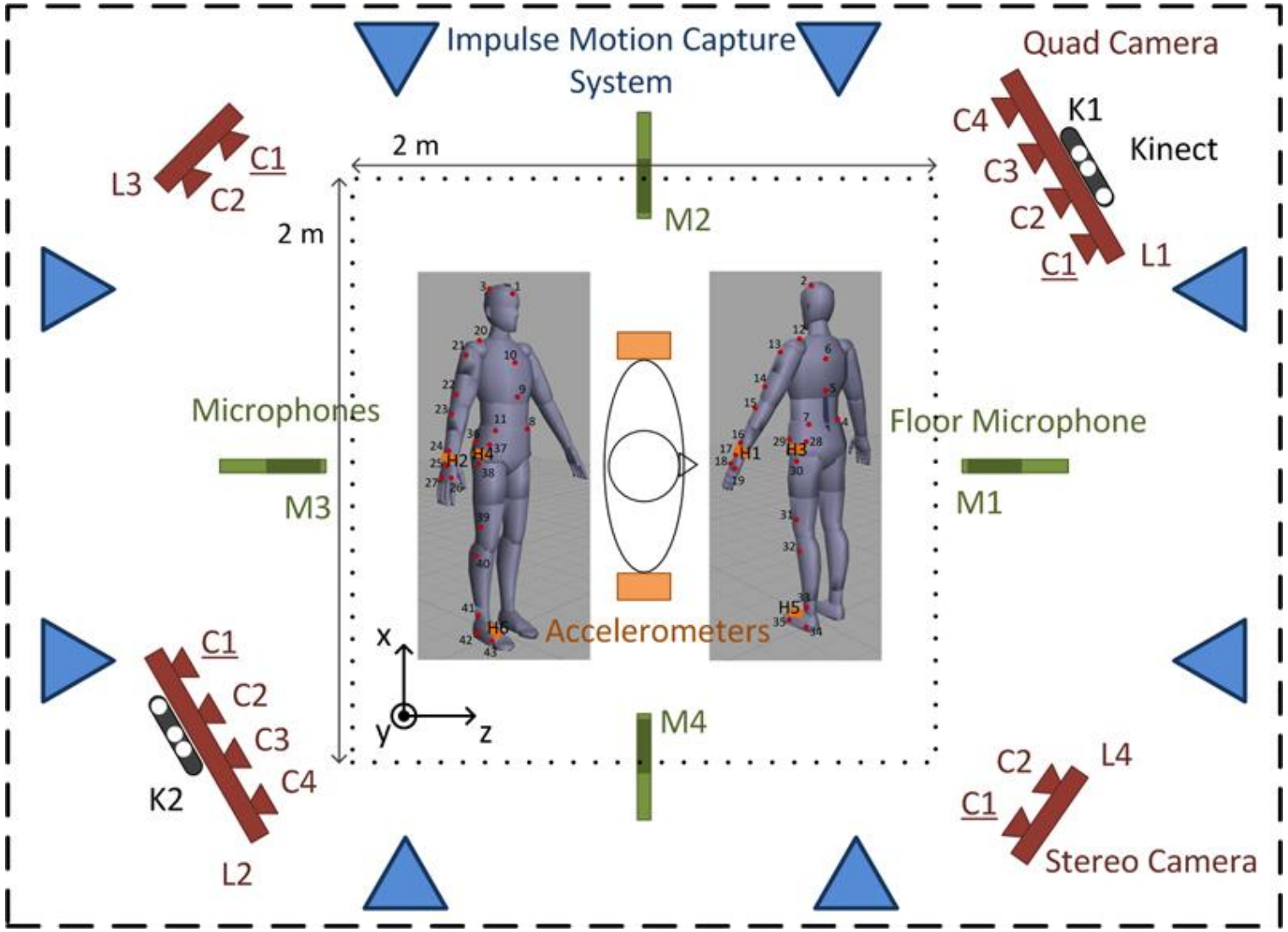




Source: http://www.utdallas.edu/~kehtar/UTD-MHAD.html

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Example: Berkeley Multimodal Human Action Database (MHAD)



Source: http://tele-immersion.citris-uc.org/berkeley_mhad

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Single modality: Audio

- Audio/sound classification: A common IoT use case for microphones is to classify the environment or activity in which the sound was captured.
- Speaker recognition: Speaker recognition, or voice recognition, attempts to determine who is speaking.
- Keyword detection: Recognize whether a particular word was uttered. Keyword recognition can also be generalized to key-phrase recognition, and both are typically used as triggers for additional activity, such as starting a session of commands or bringing up an application.
- Command and control: Refers to using a small set of phrases in speech recognition. For illustration, this could include a set of commands to control a toy car, such as “move forward,” “move backward,” “go faster,” “go slower,” “turn right/left,” etc.

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Example: Anomaly detection of time series sound data

Sound data are acquired from the SMD assembly machine at a 192 kHz sampling rate. The data collection process is shown in (a). Sequential machine operational sound data are collected from an operating SMD assembly machine, with a microphone placed as indicated by the red bounding box in (b).



(a) the Surface Mounted Device (SMD) assembly machine



(b) the microphone for collecting data

Source: Fast Adaptive RNN Encoder–Decoder for Anomaly Detection in SMD Assembly Machine, https://www.mdpi.com/1424-8220/18/10/3573

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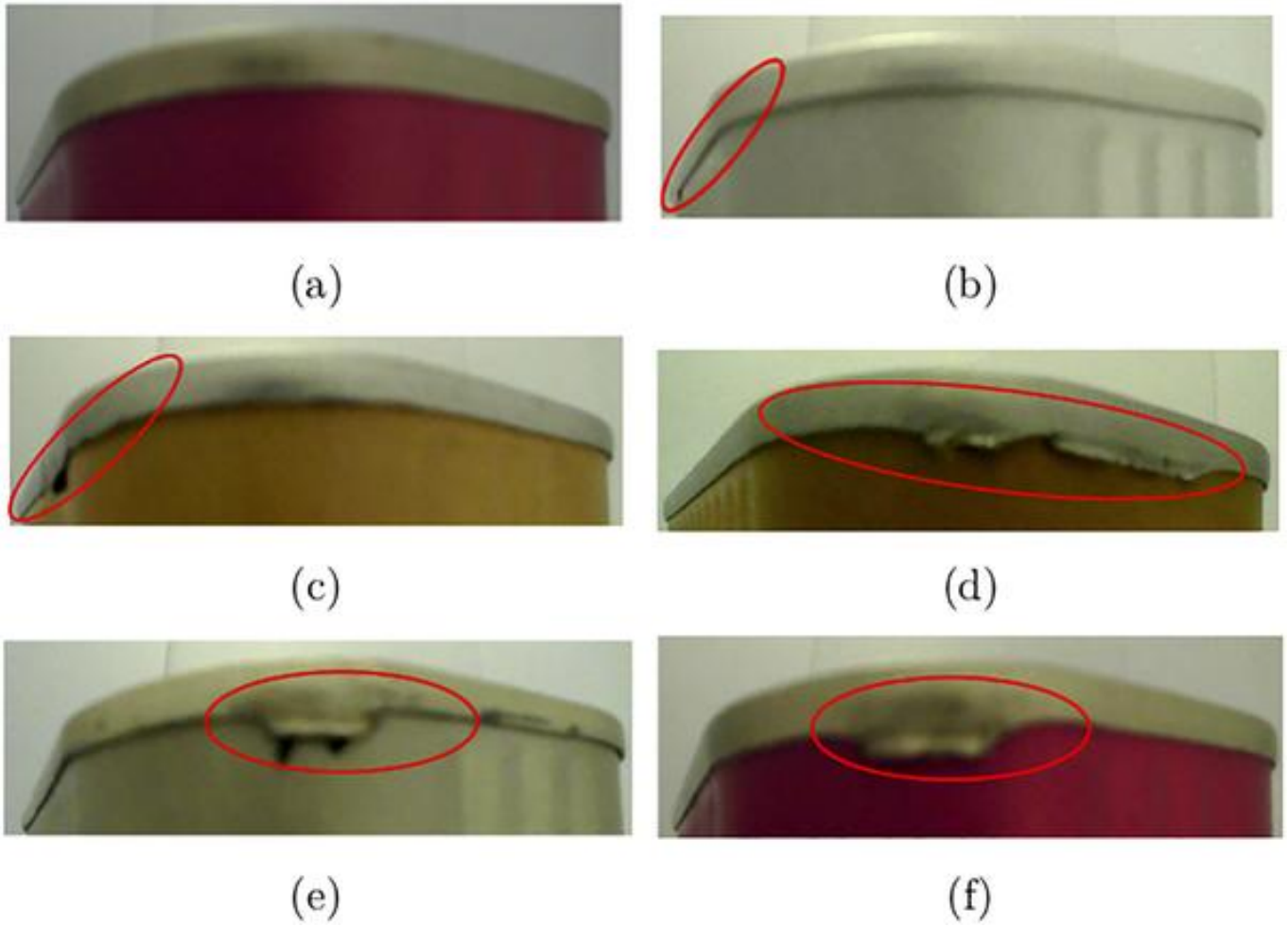
Single modality: Vision

- Scene recognition: Identify multiple objects, faces, and people in an image and use that information to determine the likely activity or context.
- Object recognition: Identify objects in an image and potentially match them to a pre-existing database of objects that have been captured before.
- Gesture recognition: Recognize static poses or moving gestures, either specific to the hand/arm or the human body.
- Anomaly detection: Identify if any anomaly occurred that should trigger additional analysis.
- Video summarization: Summarize the salient aspects of a long video stream. This includes scene changes, key scenarios, and objects/characters that are the focus of the video.

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Example: Classification of manufacturing defects

Images: (a) without defects, (b)-(c) with small defects, and (d)-(f) with big defects.

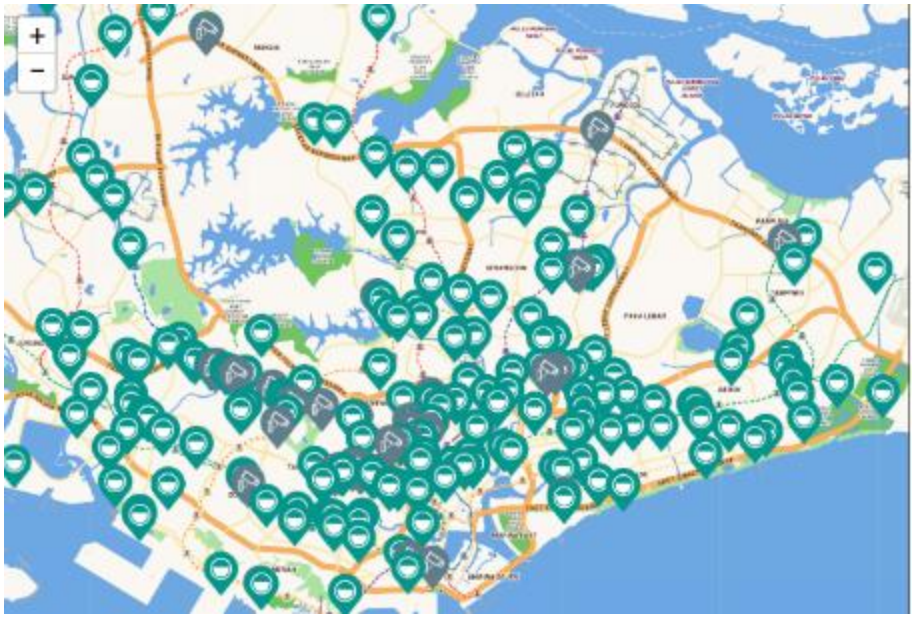


Source: Automatic detection and classification of manufacturing defects in metal boxes using deep neural networks, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0203192

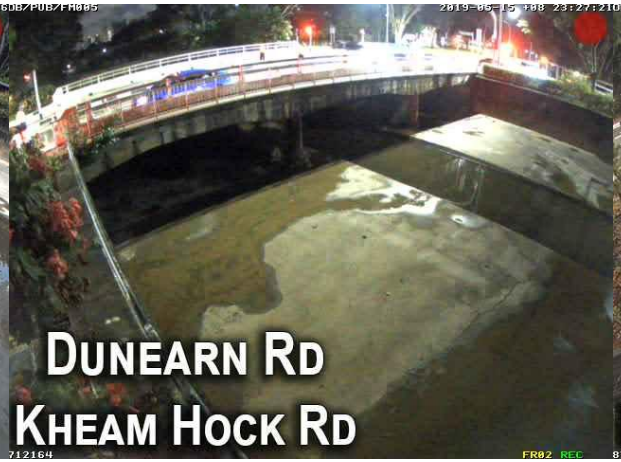
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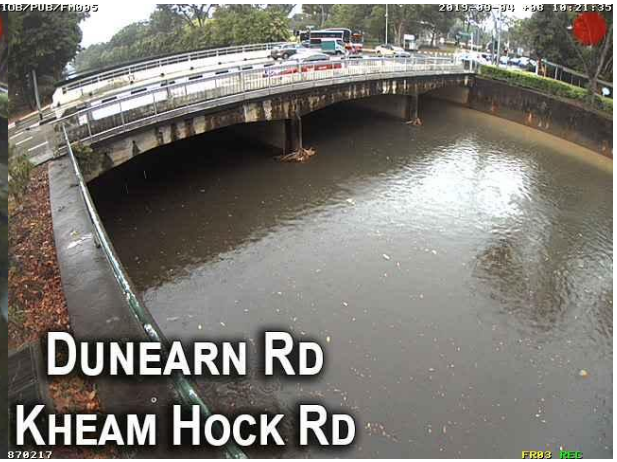
Example: Flood monitoring

- 210 water level sensors are used for monitoring the drainage system.
- 49 CCTV cameras at selected locations are used for monitoring the road.





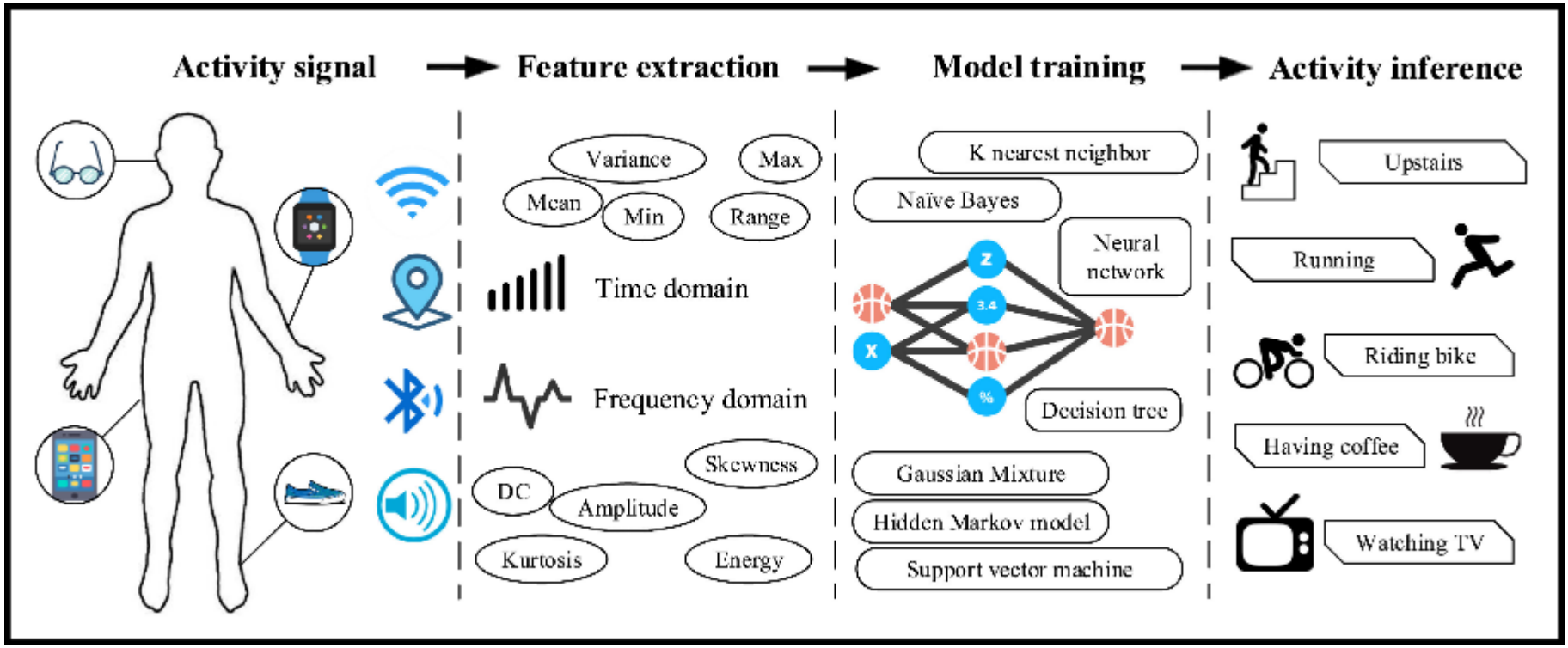




Reference: https://app.pub.gov.sg/waterlevel/pages/WaterLevelSensors.aspx

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Example: Sensor-based human activity recognition



Source: Deep learning for sensor-based activity recognition: A survey, https://arxiv.org/abs/1707.03502

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Single modality: Augmentation

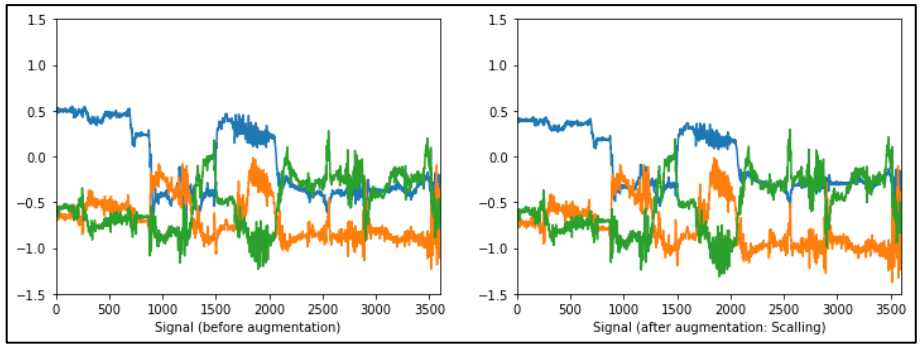
- Scaling changes the magnitude of the data in a window by multiplying by a random scalar.
- Magnitude filtering changes the magnitude of each sample by convolving the data window with a smooth curve varying around one.
- Jittering simulates additive sensor noise. These data augmentation methods may increase robustness against multiplicative and additive noise and improve performance.
- Time-warping perturbs the temporal location. By smoothly distorting the time intervals between samples, the temporal locations of the samples can be changed using time-warping.

Reference:
- Time Series Data Augmentation for Deep Learning: A Survey, https://arxiv.org/abs/2002.12478
- An empirical survey of data augmentation for time series classification with neural networks, https://arxiv.org/abs/2007.15951

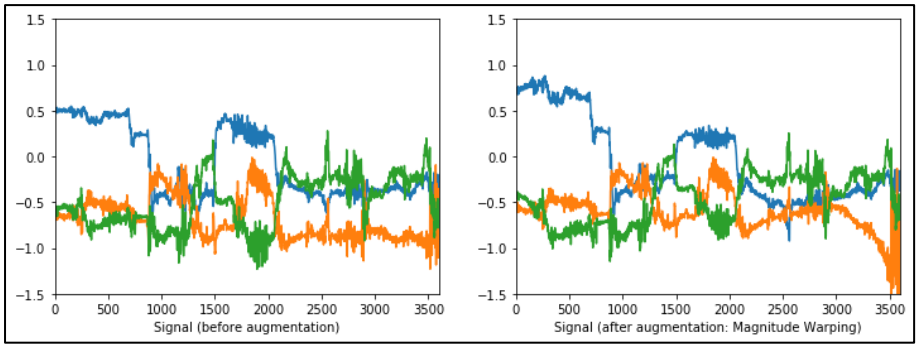
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Single modality: Augmentation

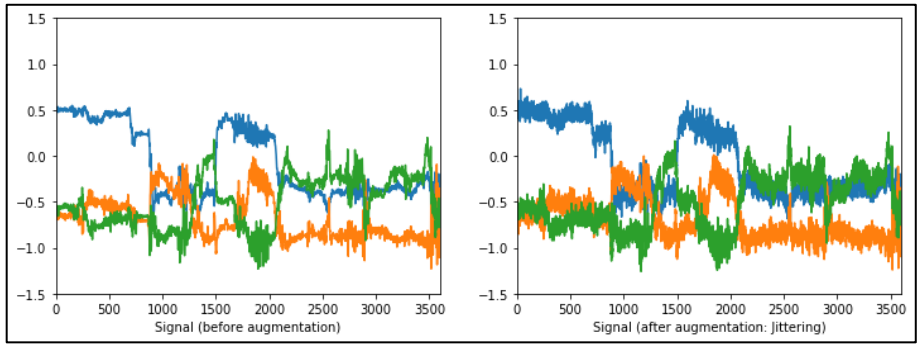
Scaling



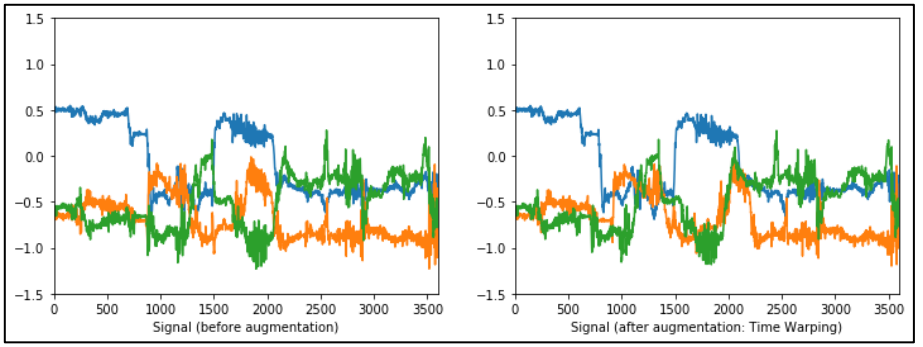
Magnitude filtering



Jittering



Time-warping



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Why multimodal (1)

- Representation: The information from the fusion process has an abstract level higher than that of each input data set.
- Accuracy: The standard deviation of the data after the fusion process is (usually) smaller than the standard deviation provided directly by the sources.
- Completeness: Bringing new information to the current knowledge on an environment allows a more complete view of this environment.

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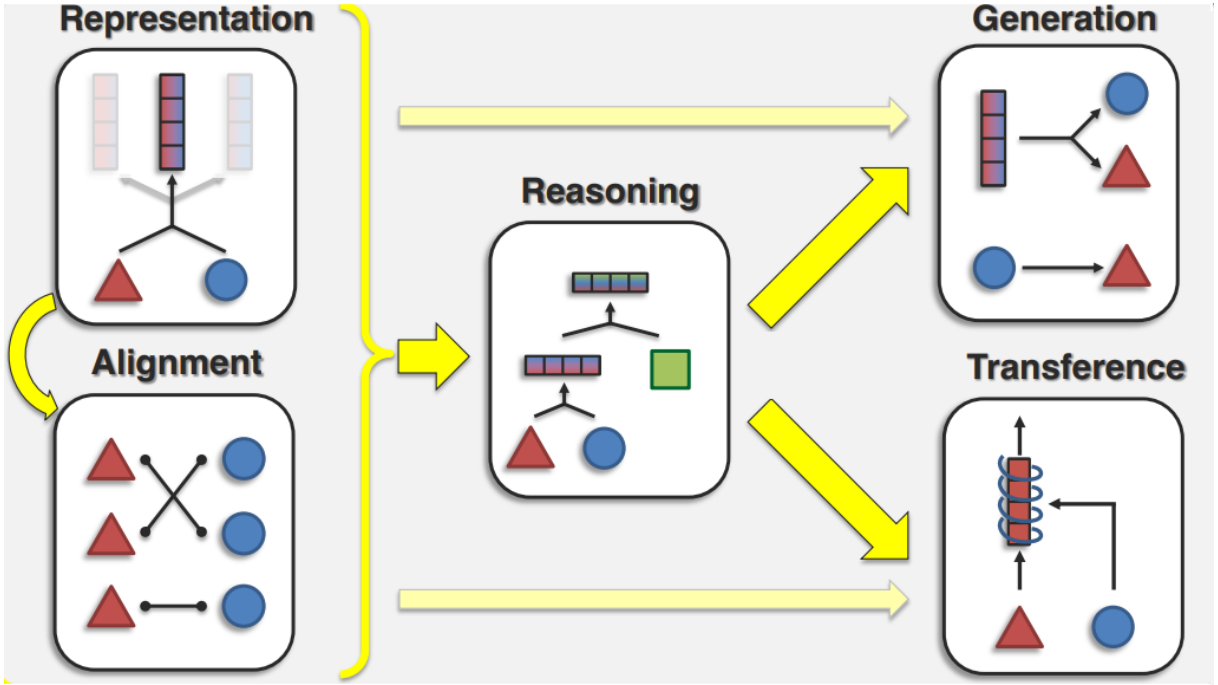
Why multimodal (2)

- Fusion across sensors (same type): Several temperature sensors measure the temperature of an object.
- Fusion across domains (same type, different definition/range): Several sensors measure the same attribute over several different ranges or domains. This arises, for example, in the definition of a temperature scale.
- Fusion across attributes (different types): Several sensors measure different quantities associated with the same experimental situation, such as measuring air temperature, pressure, and humidity to determine air refractive index.
- Fusion across time: Current measurements are fused with historical information.

Challenge: Each modality has a different quality (e.g., noisy or not), structure (e.g., spatial or temporal), and representation (e.g., discrete or continuous).

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Multimodal machine learning

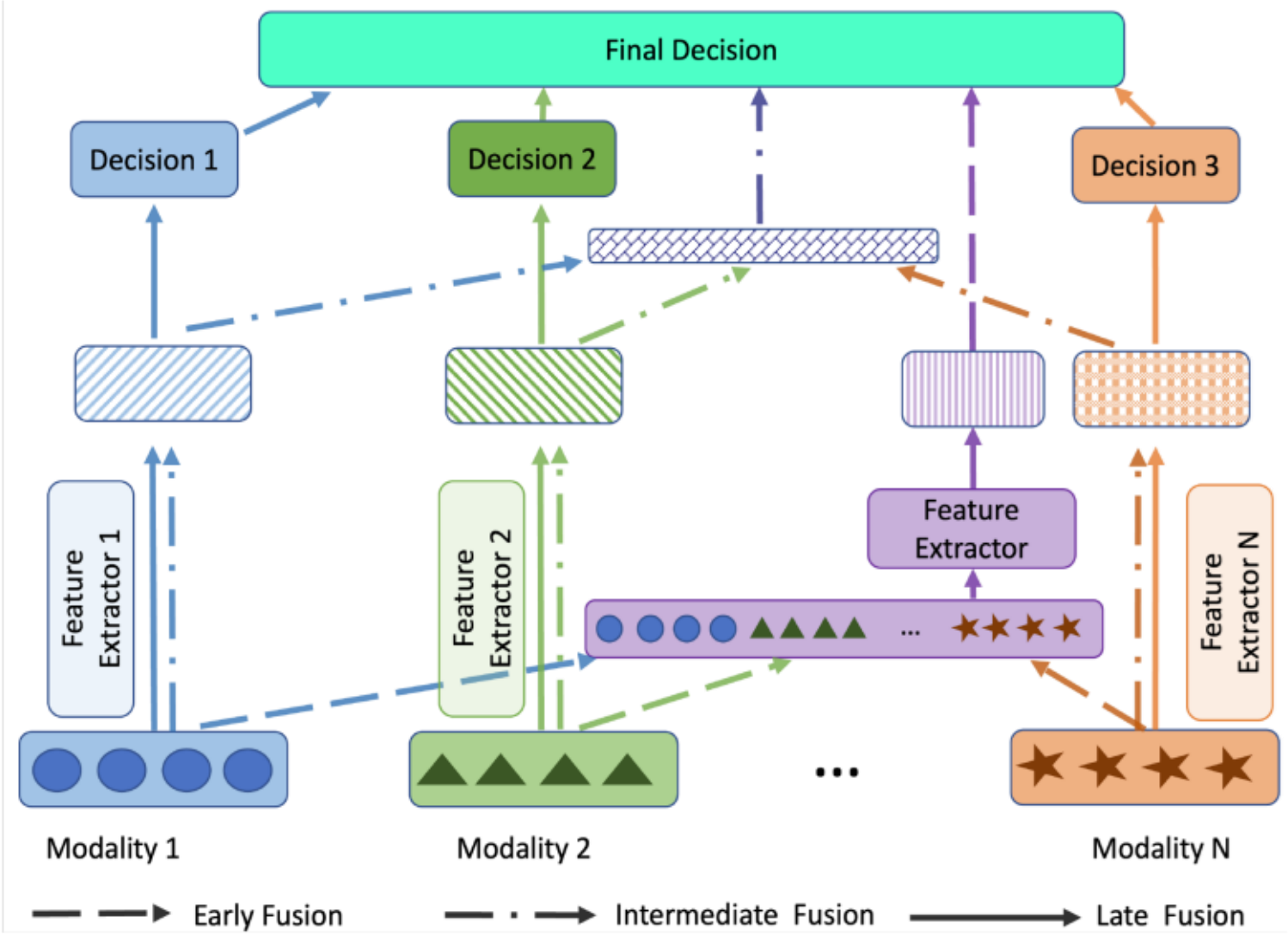


Reference: Foundations & Trends in Multimodal Machine Learning: Principles, Challenges, and Open Questions, https://arxiv.org/abs/2209.03430

- Representation studies how to represent and summarize multimodal data to reflect the heterogeneity and interconnections between individual modality elements.
- Alignment aims to identify the connections and interactions across all elements.
- Reasoning aims to compose knowledge from multimodal evidence, usually through multiple inferential steps for a task.
- Generation involves learning a generative process to produce raw modalities that reflect cross-modal interactions, structure, and coherence.
- Transference aims to transfer knowledge between modalities and their representations.

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Multimodal machine learning

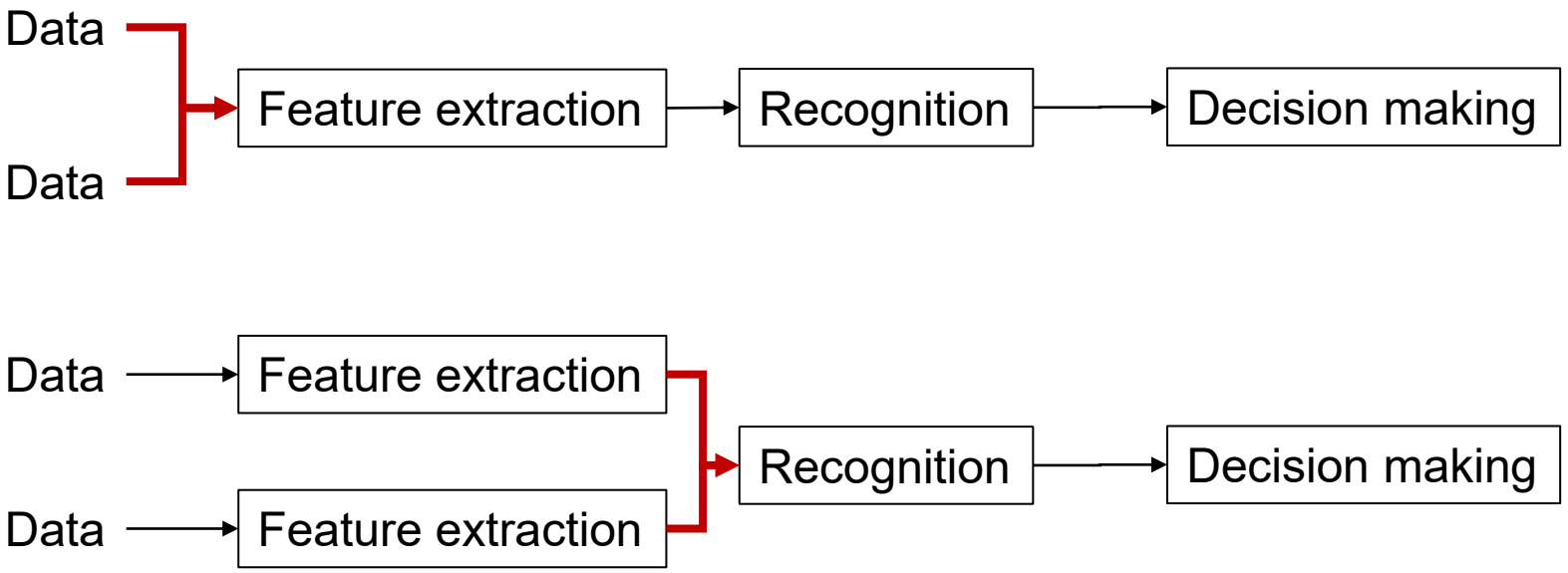


Reference: Deep Multimodal Data Fusion, https://dl.acm.org/doi/10.1145/3649447

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Multimodal sensor data representation (1)

- Tightly coupled sensor data fusion (early fusion): It allows for the combination of multiple sensors that appear as a single complex sensor for most of the applications.

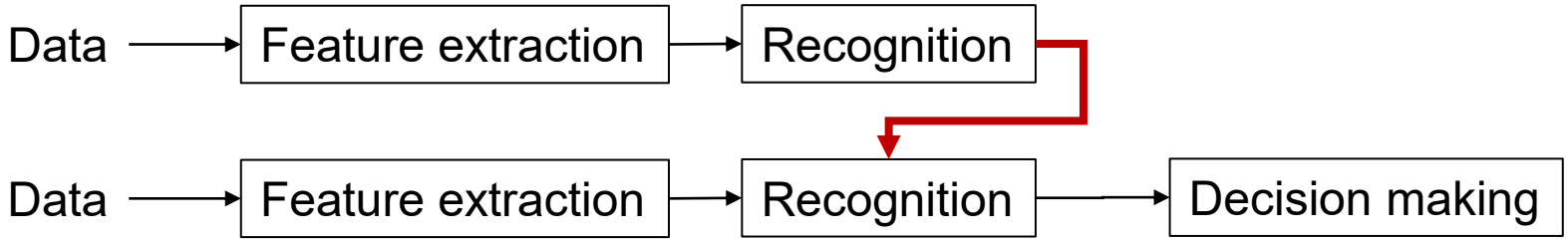


Sometimes, it is also called "intermediate fusion".

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Multimodal sensor data representation (2)

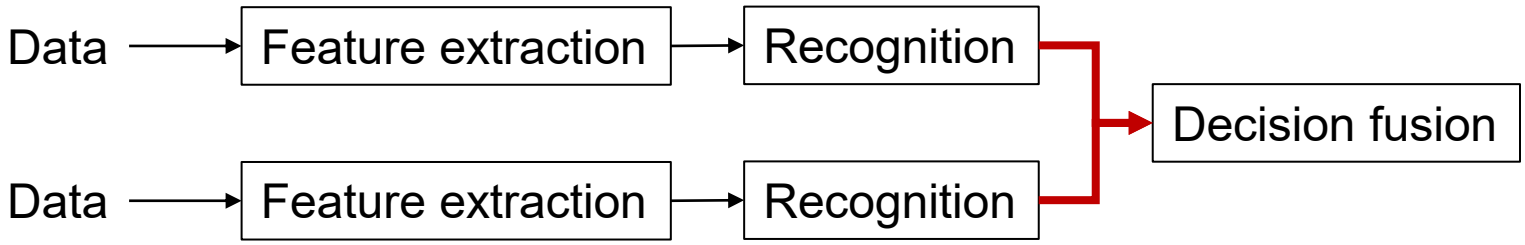
- Loosely coupled sensor data fusion (restricted fusion): Restricts the recognition search space of the mode modality based on the results from the other modality. The first recognition pipe helps to set the context for the other. Once that is accomplished, the second recognition pipe only performs recognition within the established context, leading to potential savings in computation and higher performance.



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Multimodal sensor data representation (3)

- Uncoupled sensor data fusion (late fusion): The data fusion occurs at the last possible stage in the respective pipelines. The advantage of this method is that sensor fusion is simple and can be accomplished using existing technology pipelines for recognition.



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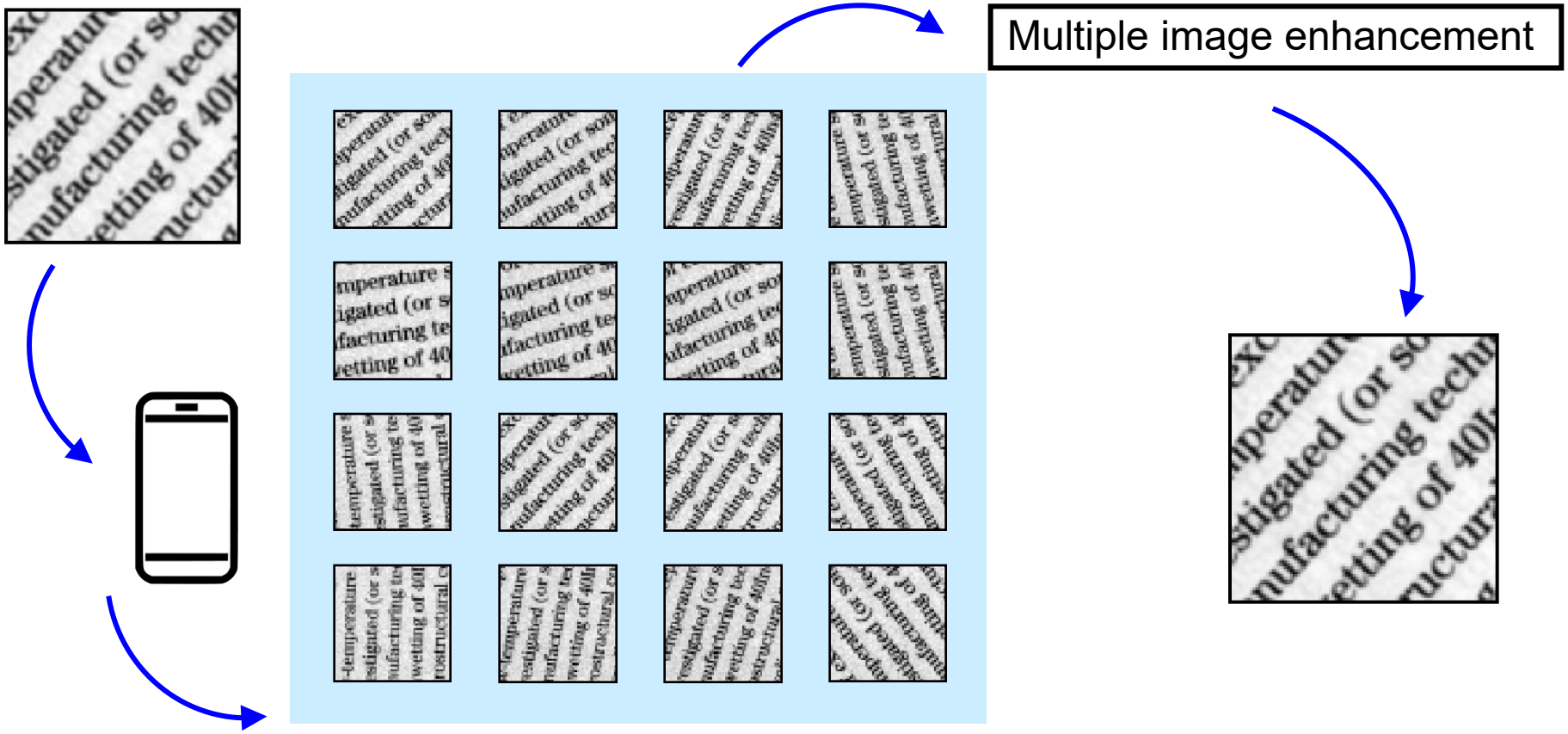
Alignment

- Spatial alignment. Transformation of the local spatial positions to a common coordinate system. The process involves georeferencing the location and field of view of each sensor.
- Temporal alignment. Transformation of the local times to a common time axis. In many applications, it is performed using a dynamic time warping algorithm.

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Alignment: Spatial

- At different times (multi-temporal fusion)



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Alignment: Temporal

Objective: Find point-to-point correspondence between two frames in the video sequence. The pixel intensities of an object do not change between consecutive frames. Neighbouring pixels have similar motion.

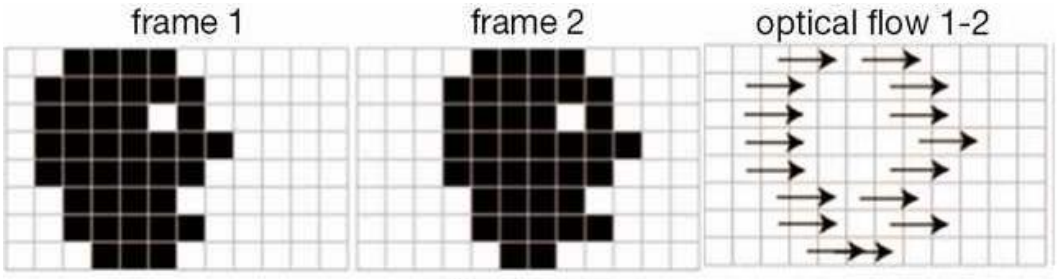






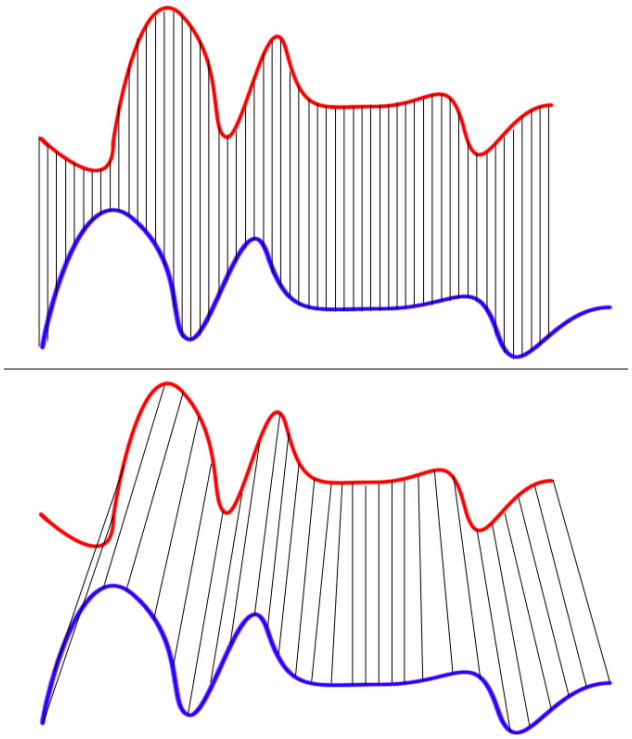
Image source: http://users.umiacs.umd.edu/~zhuolin/Keckgesturedataset.html

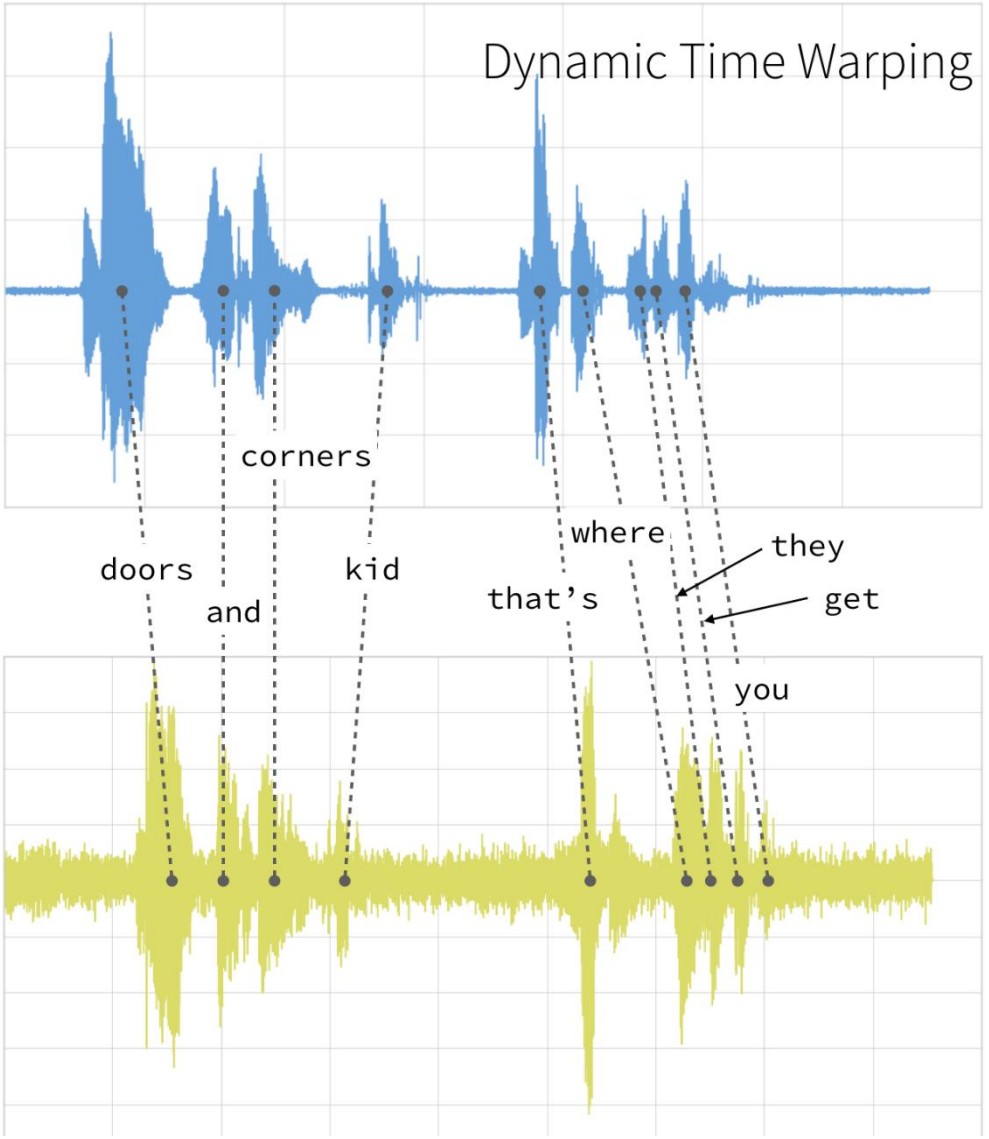
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Alignment: Temporal

Euclidean registration (top figure): Align the i-th point on one time series with the same i-th point.

Dynamic time warping registration (bottom figure): Allows similar shapes to match even if they are different in the time axis.



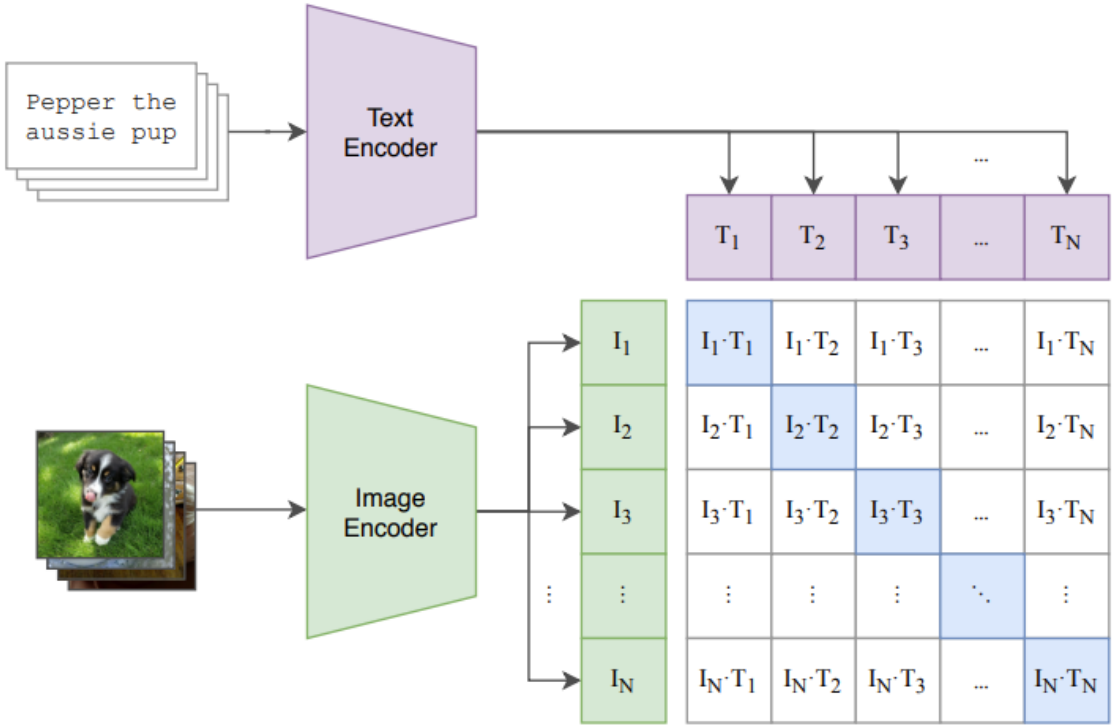


Reference: https://databricks.com/blog/2019/04/30/understanding-dynamic-time-warping.html

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Alignment: Feature (latent)

CLIP: Contrastive Language-Image Pre-training (model training)



- Objective: Train an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) examples.
- A dataset of 400 million (image, text) pairs collected on the Internet.
- Training: Given a batch of N (image, text) pairs, CLIP is trained to predict which of the $N \times N$ possible (image, text) pairings across a batch actually occurred. It jointly trains an image encoder and text encoder to maximize the cosine similarity of the image and text embeddings of the N real pairs in the batch while minimizing the cosine similarity of the embeddings of the $N^2 - N$ incorrect pairings.

Reference: Learning Transferable Visual Models From Natural Language Supervision, ICML 2021, https://arxiv.org/abs/2103.00020

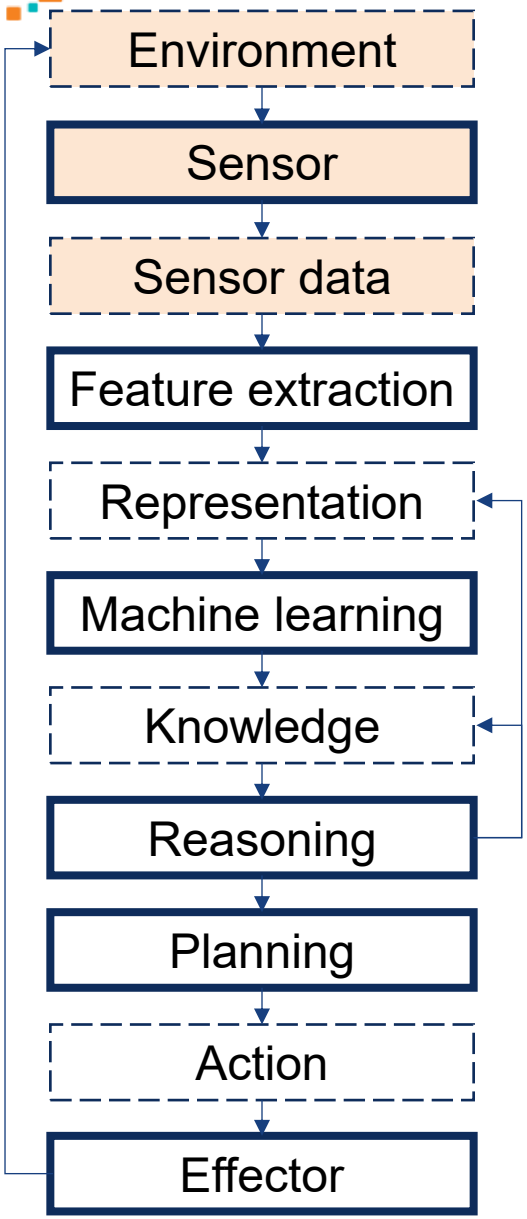
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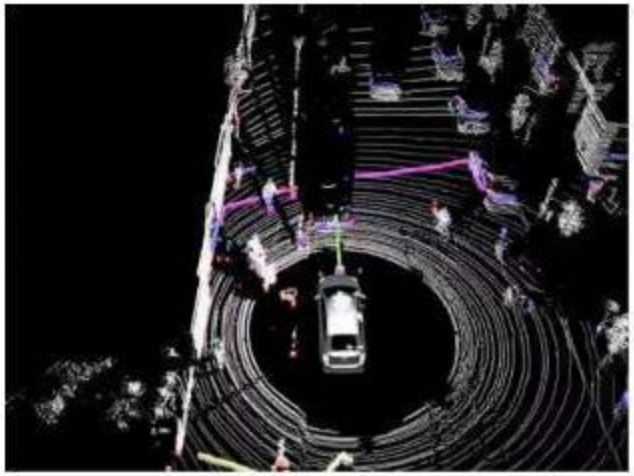


SENSE MAKING PIPELINE & CASE STUDIES

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Sense making pipeline A (1)







Lidar



Camera
(Visible, Infrared)



Radar



GPS



Stereo Camera



Microphone



Networking
(Wired, Wireless)

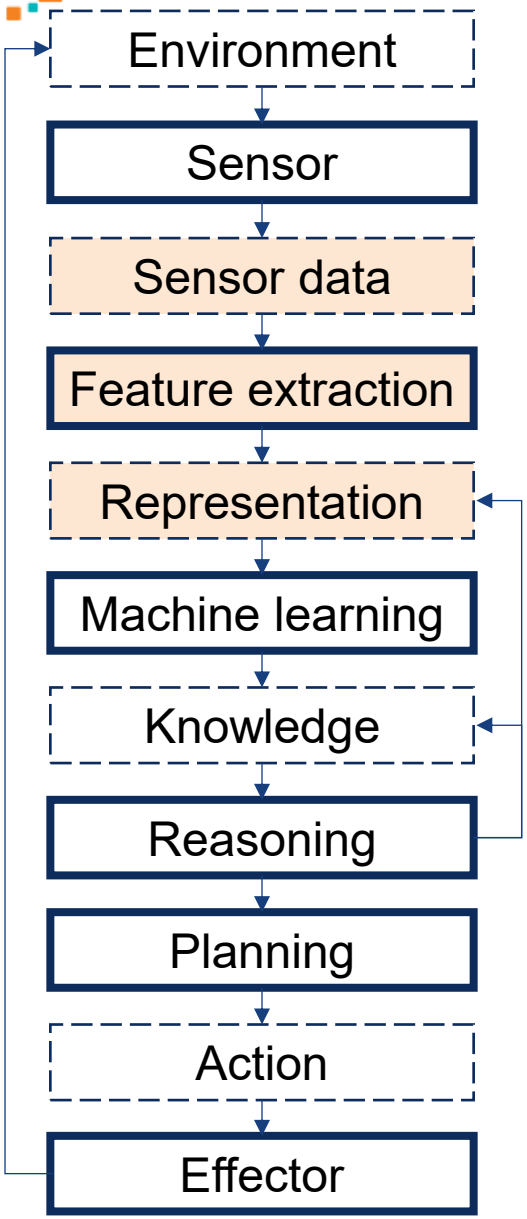


IMU

Source: MIT 6.S099: Artificial General Intelligence, https://agi.mit.edu/

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Sense making pipeline A (2)



1. Signal representation can be manually designed (input-agnostic)

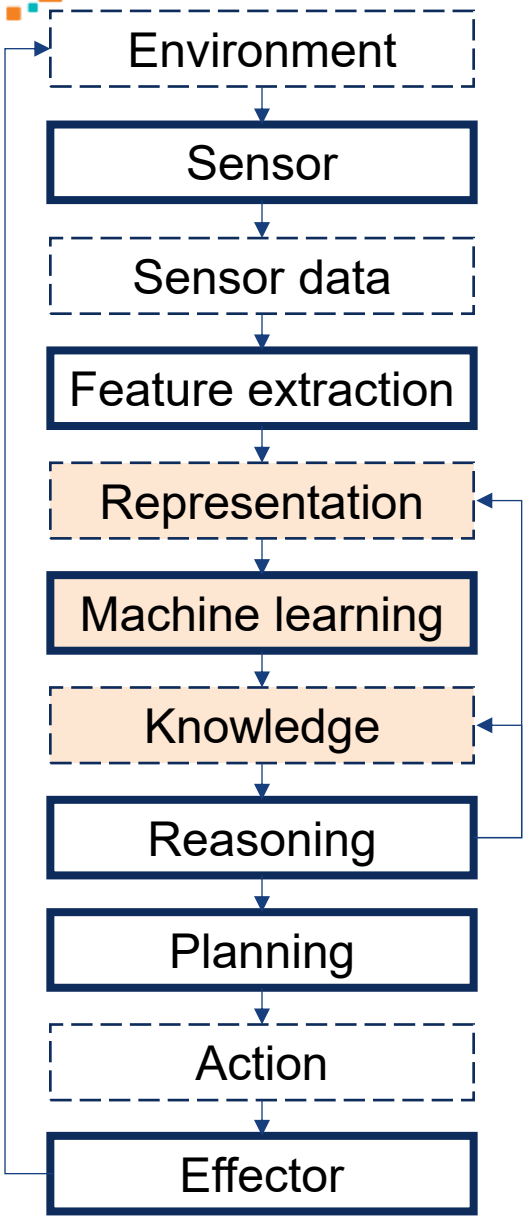
2. Signal representation can be adaptively to the input signal

3. Signal representation can be learned from signal dataset

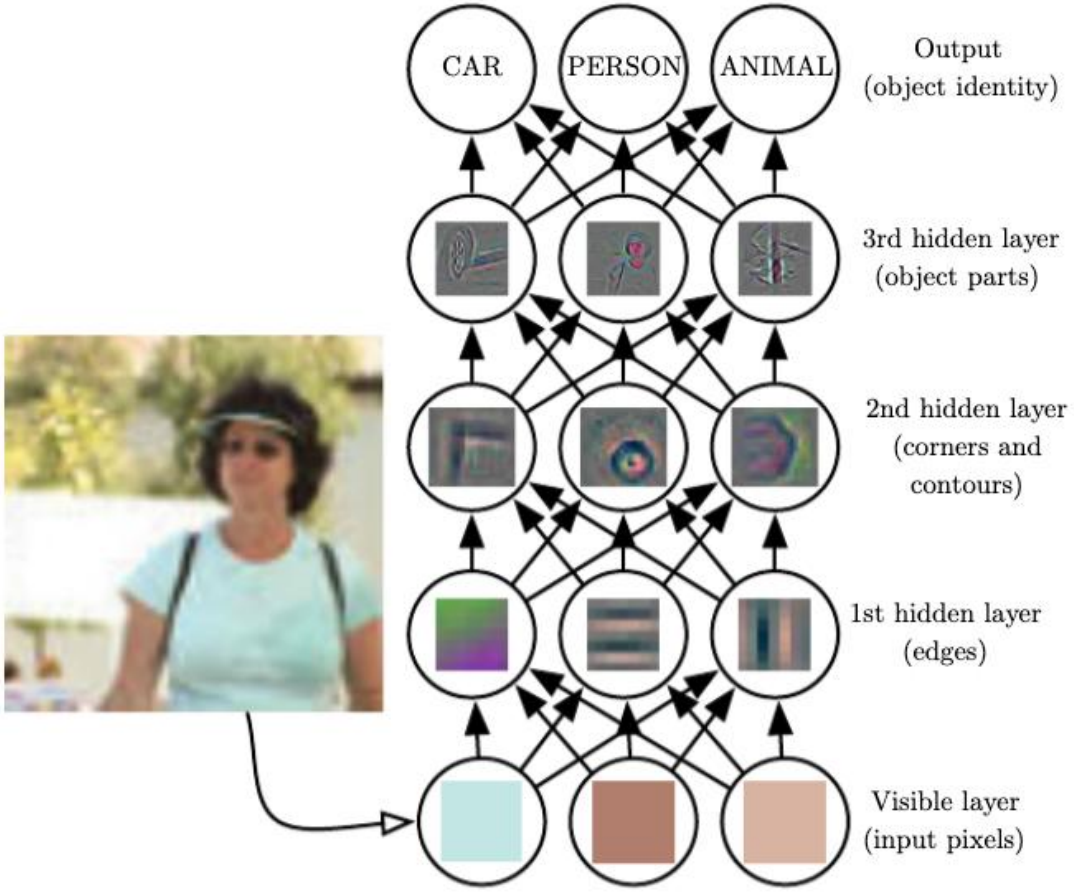
Source: MIT 6.S099: Artificial General Intelligence, https://agi.mit.edu/

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Sense making pipeline A (3)



Built machine learning model to extract knowledge from feature representation



Source: MIT 6.S099: Artificial General Intelligence, https://agi.mit.edu/

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Sense making pipeline A (4)

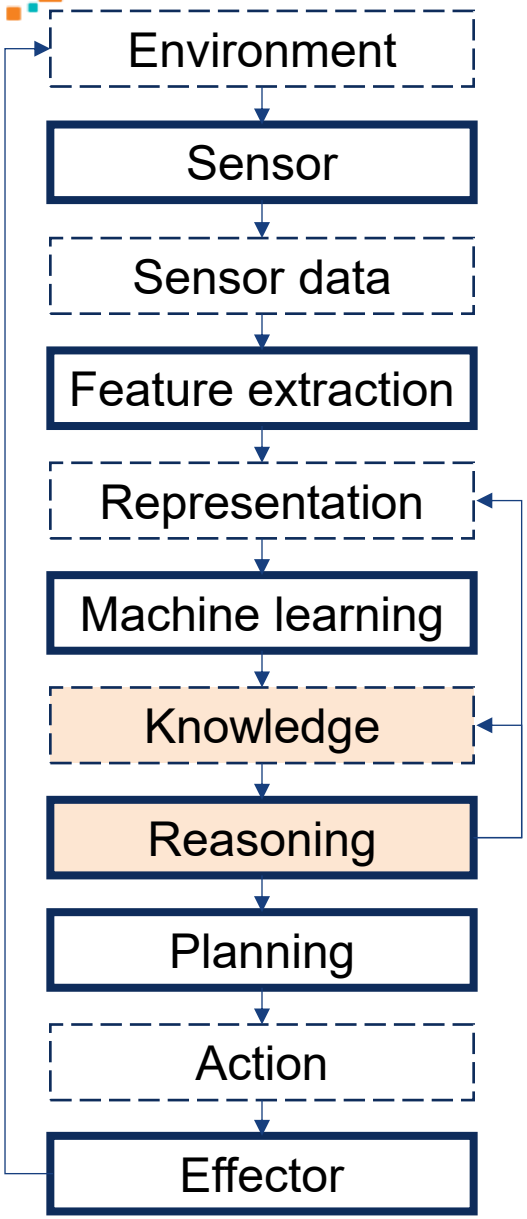


Image Recognition:
If it looks like a duck



Audio Recognition:
Quacks like a duck



Reasoning based on machine learning results.

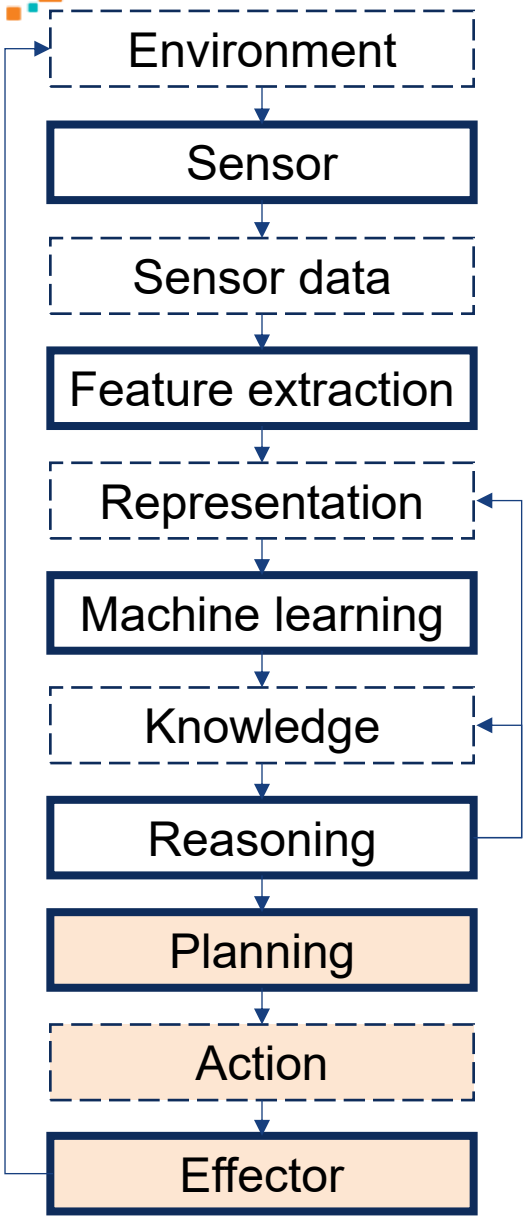
Activity Recognition:
Swims like a duck



Source: MIT 6.S099: Artificial General Intelligence, https://agi.mit.edu/

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Sense making pipeline A (5)



- Make the decision (planning)
- Implement the decision (effector)



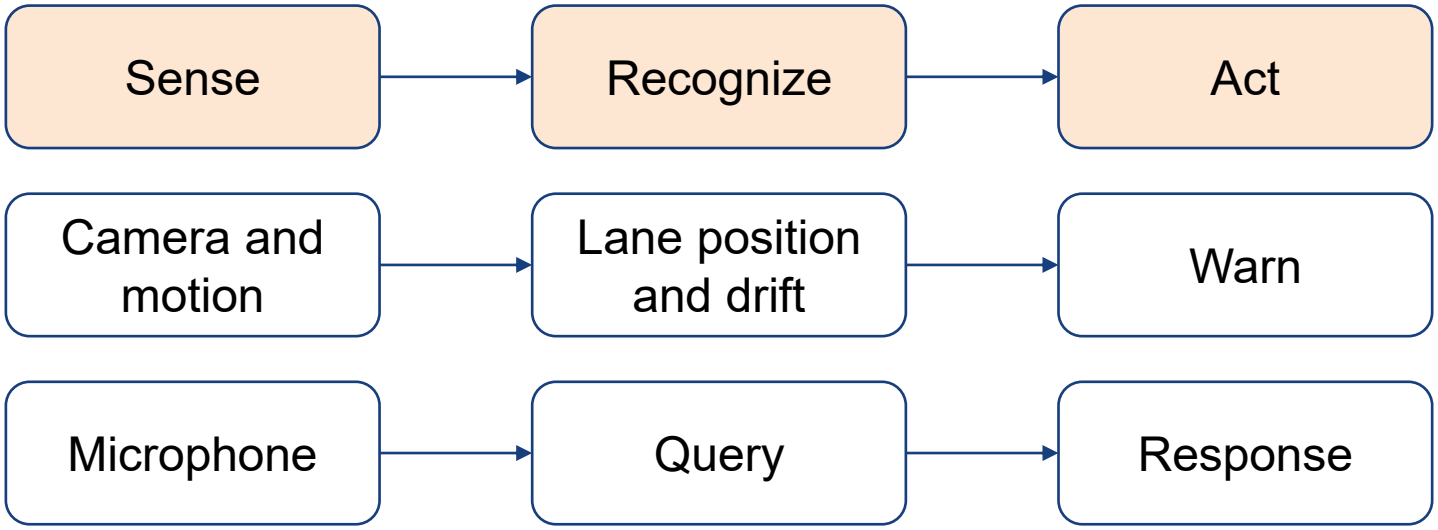
Source: MIT 6.S099: Artificial General Intelligence, https://agi.mit.edu/

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Another example

Example: Automatic Driver Assistance System (ADAS)

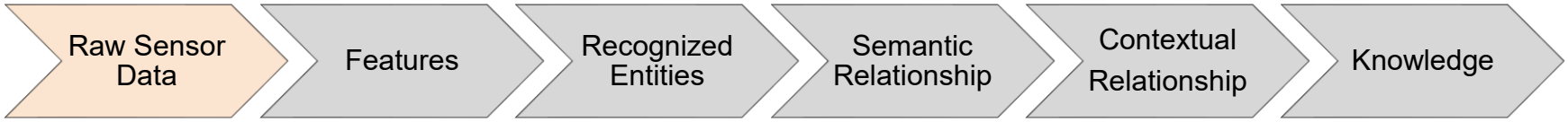
- It uses a set of sensors like cameras, motion sensors, etc., to provide a driver with assistance for safety and comfort, such as tracking the position of a car in the traffic lanes and providing audible warnings in case the car drifts from its lane (Lane Departure Warning).
- It works on the fundamental processing principles of IoT, comprising sensors collecting data (cameras, motion sensors), algorithms recognizing the data (relative position of the car in the lane, car drift), and applications acting on the recognized data (audible warning on/off).



Source: Making Sense of Sensors: End-to-End Algorithms and Infrastructure Design from Wearable-Devices to Data Centers, Apress.

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Sense making pipeline B (1)

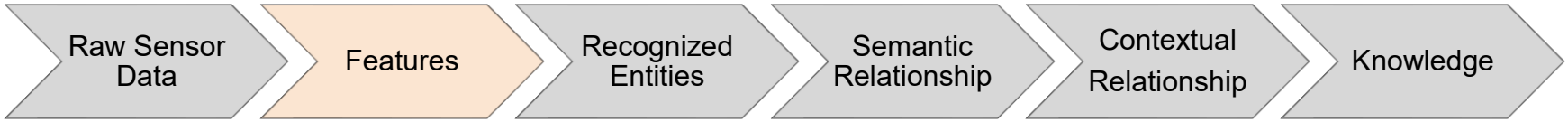


Raw Sensor Data

- Raw (or unprocessed) sensor data is captured by sensors at the front end of the pipeline
- **Vision**: Camera sensors take still or video shots of a scene
- ***Audio/Speech****:* The microphone-like sensors typically capture the audio, and analog-to-digital conversion is applied
- **Motion:** Inertial sensors typically measure acceleration and motion
- **BMI sensors**: The Brain-Machine Interface sensors, like EEG, report brain activity measurements as activity graphs
- **Other IoT sensors**: temperature, humidity, etc.

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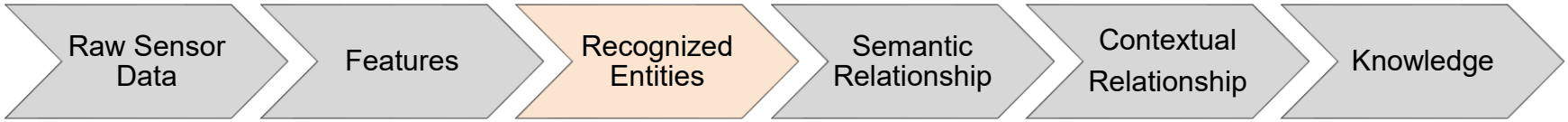
Sense making pipeline B (2)



Category	Representative features
Geometrical	Edges, lines, line widths, line relationships (e.g., parallel, perpendicular), circles, shapes, size of the enclosed area
Structural	Surface area, relative orientation, orientation in the vertical and horizontal ground plane
Statistical	Number of surfaces, area, and perimeter, mean, variance, kurtosis, skewness, entropy
Spectral	Color coefficients, spectral peaks, and lines
Time domain	Pulse characteristics (rise and fall times, amplitude), pulse width, pulse repetition interval, moments
Frequency domain	Fourier coefficients, Wavelet coefficients

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Sense making pipeline B (3)

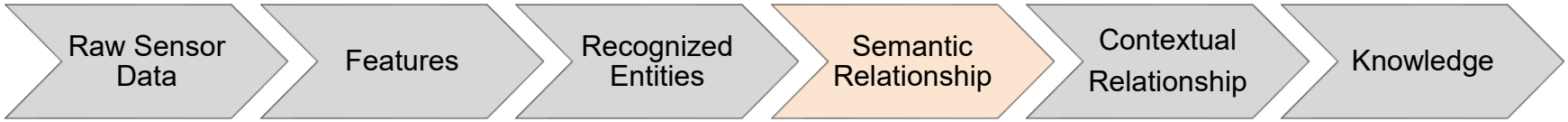


Recognized Entities

- Complex task involving spatial and temporal analysis of the extracted features to map the aggregate to predefined entities.
- For vision, the recognition task may involve classifying the extracted features to recognize shapes like objects and faces.
- For audio, statistical analysis and classification of features allow for aggregated features to be recognized as words.

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Sense making pipeline B (4)

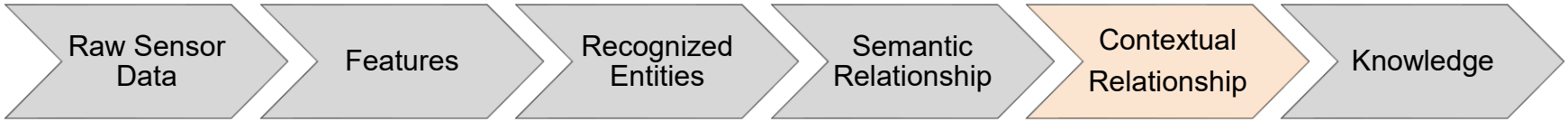


Semantic Relationship

- The connections between various recognized entities, for example, the semantic relationship between a key and a lock, refer to the operation of opening or closing the lock with the key.
- For audio, the sentence "*I like coffee*" connects "I" (the subject) with "coffee" (object) using the predicate "like."
- For vision, the semantic context of the video can be analyzed based on identified entities (multiple objects, locations, people, and activities) and scoring these entities for their co-occurrence as well as their relation to the type of classification scenario of importance.

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Sense making pipeline B (5)

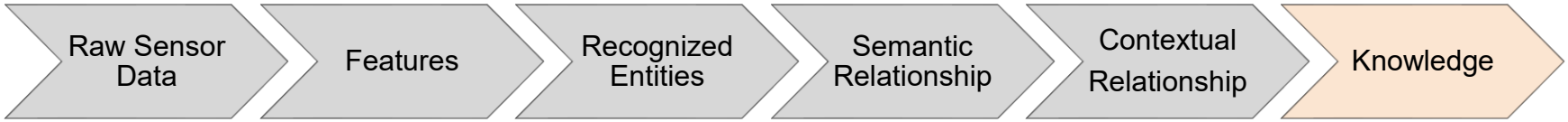


Contextual Relationship

- Understand the recognized data in context. The context can be obtained in multiple ways. These include using domain knowledge, other sensors, or the history of recognized data for context recognition.
- For vision: A face recognition surveillance system can identify a potential invasion or normal operation depending on when the recognition takes place (day vs. night) or depending on the person who is recognized.

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Sense making pipeline B (6)

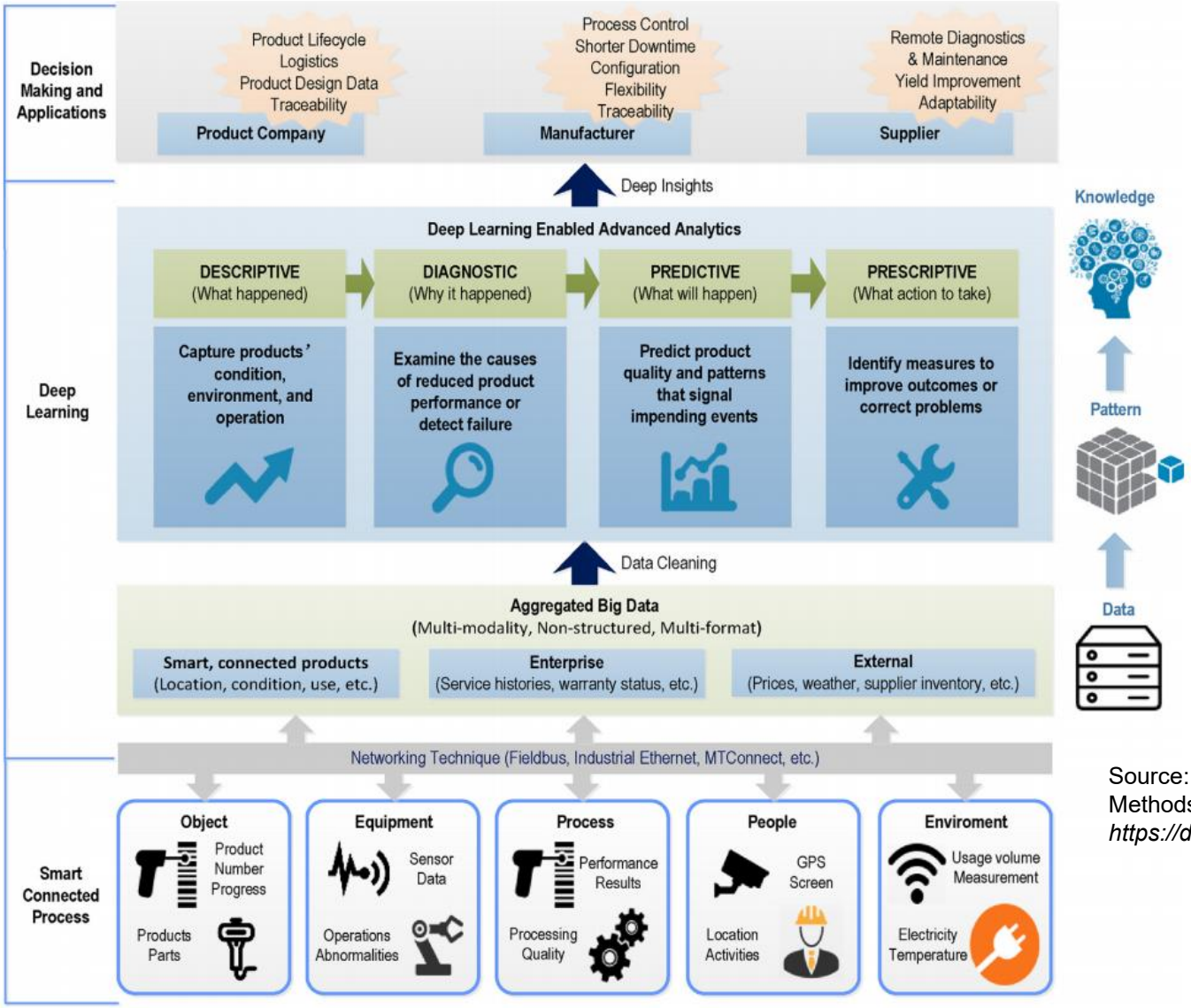


Knowledge

- Knowledge representation: Knowledge needs to be stored and represented in a manner such that the semantic information and relationships between various concepts are retained and are modifiable.
- Knowledge operations: These include tools, APIs, and programming languages to insert and extract data from the knowledge database. This involves understanding the incoming semantic data and providing a mechanism to find and manipulate the correct relationships. The extraction process involves responding to different queries targeting relationships between different entities and concepts.

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Example: Smart manufacture



Source: Deep learning for smart manufacturing: Methods and applications, *https://doi.org/10.1016/j.jmsy.2018.01.003*

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Thank you!

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